

Action-Conditioned Predictive Consistency as a Diagnostic for Gaussian-Noise Robustness in JEPA World Models

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June 2026

Abstract

Latent predictive world models avoid pixel reconstruction, but pixel-free prediction is not a robustness definition for closed-loop control. We study this gap with action-conditioned predictive consistency (ACPC), a no-retraining diagnostic for JEPA-style control checkpoints. The diagnostic combines ACPC Tail Risk (ATR), which measures high-quantile clean/noisy rollout disagreement for the same state under the same action sequence, with Selective Margin Pass Rate (SMPR), which checks that task-grounded near-boundary distinctions remain separated. Across three independent LeWM training seeds and four control tasks, no-noise checkpoints are fragile under observation-only Gaussian noise, while input-side noise training yields broad matched-Gaussian recovery bands under the specified stressor. Across the full training-noise sweep, recovered rows occupy low-ATR/high-SMPR regions, and held-out seed/task audits keep recovery-onset errors within two grid steps. These results support ATR and SMPR as bounded ACPC diagnostics for matched-Gaussian recovery bands in fixed JEPA world-model checkpoints.

Keywords: visual world models; JEPA; action-conditioned predictive consistency; visual robustness; predictive-dynamics diagnostics.

Code and data release: <https://github.com/Anguo-star/lewm-acpc-diagnostics>

1 Introduction

JEPAs predict future representations instead of reconstructing pixels, which can reduce pressure to encode high-magnitude irrelevant visual detail [1, 3, 29]. For control, however, “irrelevant” is action- and transition-dependent. A representation may ignore a background texture but must preserve a contact pose, doorway state, or object-goal relation if that distinction changes the next useful action. Encoder-level invariance is therefore an incomplete robustness target: planning uses the model’s *predictions*, so the robustness question should be asked after action-conditioned rollout.

We study this gap as a controlled diagnostic problem for JEPA latent world-model control. Let a clean history and a visually perturbed history describe the same underlying state, and evaluate both branches under the same action sequence. The desired behavior is not necessarily identical encoder latents. It is that task-relevant predicted futures agree after the shared action intervention, while states that differ in action, transition, cost, or goal relation remain separable. This selective requirement is the core of action-conditioned predictive consistency (ACPC).

The selective tension matters because consistency alone can be misleading. A collapsed model can make clean and corrupted views agree while erasing distinctions needed for planning. Conversely, a large encoder shift can be harmless if the action-conditioned predictor contracts nuisance directions before they affect the predicted rollout used by planning. The diagnostic must therefore pair a same-state predictive-stability metric with an anti-collapse margin check for task-grounded near-boundary separations.

We instantiate this diagnostic with a fixed Gaussian intervention. Across PushT, TwoRoom, Reacher, and Cube, we evaluate no-noise LeWM [19] checkpoints and full-sequence input-side Gaussian-noise checkpoints over $\sigma_{\max} \in \{0.01, \dots, 0.08\}$. The primary endpoint is observation-only Gaussian evaluation noise with the goal image kept clean. Across three independent LeWM training seeds, the no-noise baseline remains strong on clean observations but loses 74.56 ± 5.55 points on PushT and 41.00 ± 1.44 points on Reacher at evaluation noise

$\sigma = 0.08$. Test-time visual fragility is not new [15, 33, 12, 21]; our focus is what it reveals about fixed JEPA world-model checkpoints.

The evidence shows broad matched-Gaussian recovery bands rather than a stable universal point optimum. Input-side noise training recovers the observation-noise endpoint on TwoRoom, PushT, and Reacher and gives a weaker but positive Cube signal across training seeds. At recovered endpoints, ACPC Tail Risk (ATR) is substantially lower and Selective Margin Pass Rate (SMPR) is near one across tasks. Bounded blur/resize checks delimit perturbation scope.

This paper makes three contributions. First, it formulates visual robustness for JEPA world-model control as selective action-conditioned predictive consistency rather than encoder invariance. Second, it derives an ACPC radius-margin diagnostic certificate and a local Gaussian sensitivity interpretation: high-probability same-state predictive radius, planner candidate margins, and task-grounded discriminability margins jointly bound fixed-pool candidate instability and collapse risk, while the composed encoder-predictor Jacobian explains how the radius can contract under matched Gaussian training. ATR and SMPR are empirical estimates of the two sides of this certificate. Third, it provides a fixed-checkpoint Gaussian robustness study across four tasks and three LeWM training seeds, including full-sweep diagnostic trajectories, held-out diagnostic-region validation, sample-level fixed-pool event-rate recomputation, finite-difference and exact-JVP/Hutchinson Gaussian sensitivity checks, joint ATR-plus-guard audits, and bounded unseen-stressor checks that delimit perturbation scope.

2 Related Work

This diagnostic sits between latent-prediction representation learning and robust visual-control methods. The related work is organized around three questions: what latent prediction promises, where augmentation-based invariance is insufficient for control, and how control-relevant representation work treats downstream predictive consequences.

2.1 Latent prediction and JEPA world models

JEPA [16] predicts in latent space rather than reconstructing pixels; I-JEPA [1] and the V-JEPA family [3, 2, 20] extend this idea to images, video, and dense physical-world representations. LeWM [19], our baseline, stabilizes an end-to-end JEPA world model with SIGReg [6]. PLDM [25, 26], via `stable-worldmodel` [18], provides related latent-dynamics planning context. LeJEPA theory [14] studies latent-variable recovery and latent planning; our diagnostic asks whether learned visual world-model checkpoints give paired clean/noised views consistent action-conditioned predictions.

2.2 Invariance, augmentation, and visual robustness

Augmentation and joint-embedding losses impose invariances only when their training signal identifies the right nuisance factors [31, 27, 35]. Recent theory clarifies why latent prediction can help with nuisance or noisy features but also why it is not enough for control. Van Assel et al. [29] and Littwin et al. [17] analyze representation-level effects; they do not address closed-loop action-conditioned rollout, candidate-cost stability, or the need to keep action-distinct states separable. Seq-JEPA [8] addresses a related invariance-equivariance tension architecturally.

We do not train robust visual-control systems as competing baselines. DrQ/DrQ-v2 [15, 33], SODA/DMC-GB [12], DreamerV3 [10], TD-MPC2 [11], and ViGMO [21] define future method-comparison axes; noisy/video JEPA variants [22, 13, 23] are representation-level context. ACPC is complementary: it tests whether same-state visual perturbations become equivalent after a fixed action intervention while action-, transition-, or cost-distinct cases remain discriminable.

2.3 Control-relevant predictive representations

Bisimulation merges states with equivalent reward and transition behavior. DeepMDP [7], DBC [34], bisimulation-regularized MPC [24], and Bisim-JEPA [28] use this principle to learn control-relevant representations. Value-equivalent and value-aware models [9, 30] and VIBR [5] likewise emphasize downstream control consequences. Wang et al. [32] formalize action-conditioned video world modeling as a group action with identity, inverse, and composition-consistency metrics. ReOI [4] treats visual distractor robustness in visual MPC through observation interventions for world-model action-outcome prediction. These lines show that downstream

predictive consequences are already central to control-representation work. Our diagnostic instantiates that view for paired clean/noised JEPA latent-world-model rollouts: same-state noised and unperturbed views should agree after the same action sequence, while action-distinct transitions remain separable.

Our use of ACPC is therefore narrower than an identifiability theorem, a learned bisimulation metric, or a robust MPC intervention. LeJEPA-style results motivate latent prediction and planning under latent recovery assumptions; bisimulation and value-aware models define task abstractions; group-action metrics test action faithfulness; ReOI modifies observation handling for robust visual MPC. This paper instead supplies a JEPA checkpoint diagnostic centered on additive Gaussian pixel-noise stress tests: predictive stability is tested on paired same-state clean/noised rollouts under the same action sequence, fixed-candidate cost stability is derived as a sufficient condition, and a separate discriminability guard checks that action-, transition-, or cost-distinct cases have not been collapsed.

3 Action-Conditioned Predictive Consistency

This section defines action-conditioned predictive consistency and its discriminability guard. We use ACPC to separate two requirements that encoder invariance alone conflates: same-state visual perturbations should agree after action-conditioned rollout, while task-grounded different-state pairs should remain separable. Section 3.3 then maps these two requirements to ATR and SMPR.

3.1 Setup and definitions

Let o_t be an unperturbed observation and $\tilde{o}_t = \tau(o_t)$ a visually perturbed view of the *same* underlying state (τ is additive Gaussian pixel noise here, but the definitions are perturbation-agnostic). Let h_t and $\tilde{h}_t$ be the corresponding observation-action histories, E_θ the encoder, and F_θ the action-conditioned latent predictor. Fix an action sequence $\mathbf{a}_t^H = a_{t:t+H-1}$ that is *identical* on both branches. For compact notation below, $\mathbf{a}_{0:k-1}$ denotes the length- k prefix of this sequence after re-indexing time at t , and we write

$$z_t = E_\theta(h_t), \quad \tilde{z}_t = E_\theta(\tilde{h}_t), \quad \hat{z}_{t+k} = F_\theta^k(z_t, \mathbf{a}_{0:k-1}), \quad \hat{\tilde{z}}_{t+k} = F_\theta^k(\tilde{z}_t, \mathbf{a}_{0:k-1}). \quad (1)$$

Let Π denote a *task-relevant predictive projection* — the full latent, a transition delta, goal-distance or cost features, or a learned projection — and let d be a distance on its range.

Action-conditioned predictive consistency. For a same-state perturbation pair under the shared action sequence,

$$\text{ACPC}_H(o_t, \tilde{o}_t, \mathbf{a}) = \sum_{k=1}^H \alpha_k d(\Pi(\hat{z}_{t+k}), \Pi(\hat{\tilde{z}}_{t+k})), \quad \alpha_k \geq 0. \quad (2)$$

For the theoretical statements below, we use the induced horizon metric

$$d_H((u_1, \dots, u_H), (v_1, \dots, v_H)) = \sum_{k=1}^H \alpha_k d(u_k, v_k), \quad (3)$$

so that Equation (2) is exactly the clean/perturbed projected-rollout distance under the shared action sequence. In this diagnostic framing, robustness corresponds to small ACPC_H together with action-relevant discriminability, not to the encoder distance $d(z_t, \tilde{z}_t)$ alone. Encoder geometry remains an indispensable first-stage risk signal: noisy views should stay in the same-state predictive basin and avoid crossing into task-distinct neighborhoods. The narrower point is that raw encoder distance alone is not a complete robustness criterion, because the effect of an encoder shift depends on local action-conditioned rollout sensitivity and candidate margins.

Action-relevant discriminability. Consistency must be *selective*. Let $Y^H(s, \mathbf{a})$ denote a task-relevant future signal that is external to the learned predictor — simulator state, cost-to-go, inverse-dynamics label, contact/keyframe tag, or another diagnostic proxy available in the dataset. For two genuinely different states s_i, s_j (with latents z_i, z_j) that admit an action sequence under which this external future signal diverges,

$$\Delta_{\text{dyn}}^H(s_i, s_j, \mathbf{a}) = d_Y(Y^H(s_i, \mathbf{a}), Y^H(s_j, \mathbf{a})) > m, \quad (4)$$

the representation and predictions should keep them separable,

$$d(\Psi(z_i), \Psi(z_j)) > m', \quad (5)$$

where Ψ is a discriminability projection, possibly action-conditioned (latent, predicted delta under $\mathbf{a}$, inverse-dynamics feature, cost feature, or rollout embedding), and $m, m' > 0$ are margins. Equations (2)–(5) state the target: lower predictive disagreement for same-state visual perturbations must preserve discriminability among state/action/transition-distinct pairs. A method that drove $\text{ACPC}_H \rightarrow 0$ by collapsing the latent would violate (5), so the two conditions are inseparable.

The same guard can be posed over state–action pairs: if two pairs have external task-relevant futures that differ by more than a margin, their projected rollouts should remain separated. This includes the same clean state evaluated under two dynamically distinct actions. The reported SMPR instantiates the guard with task-grounded near-boundary state pairs rather than adding a separate action-faithfulness metric.

3.2 Predictive tubes and task margins

The selective condition can be written as a radius–margin event. For a fixed action sequence, define the projected rollout map

$$G_{\mathbf{a}}(z) = \Pi(F_{\theta}^{1:H}(z, \mathbf{a})) \quad (6)$$

and the same-state predictive radius under perturbation level σ as

$$R_{\sigma}(h, \mathbf{a}) = d_H(G_{\mathbf{a}}(E_{\theta}(h)), G_{\mathbf{a}}(E_{\theta}(\tilde{h}_{\sigma}))) , \quad r_{1-\alpha}(\sigma) = Q_{1-\alpha}(R_{\sigma}(h, \mathbf{a})). \quad (7)$$

The radius side asks whether a visual perturbation remains inside the same-state predictive tube after the action intervention. The margin side asks whether planner and task-grounded distinctions remain outside that tube. For a fixed candidate pool $\mathcal{A}$, the clean planner margin is

$$\Delta_{\mathcal{A}}(h, g) = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g), \quad (8)$$

where lower cost is preferred and $\mathbf{a}^{(1)}, \mathbf{a}^{(2)}$ are the clean top-1 and top-2 candidates. For task-grounded different-state pairs, we write the discriminability margin as

$$M_{\text{diff}}(i, j, \mathbf{a}) = d(\Psi(z_i), \Psi(z_j)), \quad (9)$$

and track the failure event $M_{\text{diff}} \leq R_{\sigma} + \delta_m$ for a reporting margin $\delta_m \geq 0$. Robustness requires both sides: small radius without margin is collapse, while large margin without small radius is visually unstable planning.

3.3 Operational consistency diagnostics

The reported diagnostic uses two metrics matched to this radius–margin logic. ACPC Tail Risk (ATR) estimates the upper quantile of the same-state predictive radius in Equation (7): it summarizes normalized clean/noisy ACPC-H rollout disagreement under a fixed recorded action sequence; lower is better. We use the 90th percentile in all experiments as a fixed empirical reporting choice, not a theoretical constant. Selective Margin Pass Rate (SMPR) estimates the margin-preservation side: it is the pass rate of task-grounded label-crossing pairs whose clean different-state rollout distance exceeds the same-state noisy radius at margin zero; higher is better.

The diagnostic gate is therefore not ATR alone. ATR tracks the high-tail term $\Pr[D > \epsilon]$ in the sampled-pool tail argument, while SMPR estimates whether task-grounded near-boundary pairs remain outside the perturbation tube. Low ATR without high SMPR is not interpreted as robustness, because a collapsed encoder–predictor can also make clean and corrupted views agree. Encoder geometry and planner-side audits remain useful development checks, but the diagnostic used in this paper is the joint radius–margin event.

3.4 ACPC as a fixed-candidate stability condition

ACPC is connected to planning through the candidate costs evaluated by model-predictive control. Let h be a clean history, $\tilde{h}$ the corresponding perturbed history, and let both branches evaluate the same candidate action sequence $\mathbf{a}$. For a goal or task descriptor g , define

$$C_h(\mathbf{a}, g) = J(\Pi(\hat{z}_{t+1}), \dots, \Pi(\hat{z}_{t+H}), g), \quad (10)$$

where J is the planner cost on the projected-rollout sequence. $C_h(\mathbf{a}, g)$ is defined analogously from the perturbed branch. The following statements are sufficient conditions for stability on a fixed candidate set; they do not prove stability of the full CEM sampling distribution or of the closed-loop trajectory over repeated replanning.

Encoder geometry enters the planning link through the composed rollout map

$$G_{\mathbf{a}}(z) = \Pi(F_{\theta}^{1:H}(z, \mathbf{a})).$$

Smaller same-state encoder shifts and reduced neighborhood crossing lower upstream risk, but the planning-relevant quantity is the projected-rollout distance after the shift is passed through $G_{\mathbf{a}}$. The empirical diagnostic therefore reports action-conditioned rollout-disagreement tails rather than raw encoder distances alone.

If the planner cost J is locally L_J -Lipschitz on a fixed clean/perturbed projected-rollout neighborhood, then a paired rollout discrepancy at most ϵ implies candidate-cost drift at most $L_J\epsilon$:

$$d_H(\Pi(F_{\theta}^{1:H}(E_{\theta}(h), \mathbf{a})), \Pi(F_{\theta}^{1:H}(E_{\theta}(\tilde{h}), \mathbf{a}))) \leq \epsilon \quad \Rightarrow \quad |C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J\epsilon.$$

For a shared candidate set, if every candidate cost moves by at most η and the clean top-1/top-2 margin Δ is larger than 2η , the top candidate is unchanged. Equivalently, the ACPC-cost condition gives fixed-candidate stability when $\Delta > 2L_J\epsilon$. These are fixed-candidate statements only; adaptive sampling, repeated replanning, and environment feedback can still change the closed-loop trajectory. Formal statements and proofs are in Appendix A.

Theorem 1 (ACPC radius–margin certificate). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\} \sim q^K$ be a once-sampled candidate pool with deterministic tie-breaking. For a clean history h and matched perturbed history $\tilde{h}$, define*

$$D_j = d_H(G_{\mathbf{a}^j}(E_{\theta}(h)), G_{\mathbf{a}^j}(E_{\theta}(\tilde{h}))).$$

Assume the planner cost J is locally L_J -Lipschitz on the evaluated projected-rollout neighborhood and that for a single sampled candidate

$$\Pr_{\mathbf{a} \sim q}[D(\mathbf{a}) > \epsilon] \leq \alpha.$$

Let $\Delta_{\mathcal{A}} = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g)$ be the clean top-1/top-2 candidate margin. Then

$$\Pr_{\mathcal{A}} \left[\arg \min_j C_h(\mathbf{a}^j, g) \neq \arg \min_j C_{\tilde{h}}(\mathbf{a}^j, g) \right] \leq K\alpha + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon].$$

If the task-grounded discriminability failure event satisfies

$$\Pr[M_{\text{diff}} \leq R_{\sigma} + \delta_m] \leq \gamma,$$

then the selective-ACPC diagnostic failure risk is controlled by

$$\Pr[\text{selective failure}] \leq K\alpha + \Pr_{\mathcal{A}}[\Delta_{\mathcal{A}} \leq 2L_J\epsilon] + \gamma.$$

This is a diagnostic certificate for a fixed candidate pool and a matched perturbation distribution. It says that a checkpoint is in the safe radius–margin regime only when the high-probability same-state predictive radius is below the relevant planner margin and task-grounded separations remain outside the perturbation tube.

Definition 1 (Matched-perturbation diagnostic region). *For a checkpoint family indexed by training perturbation level ρ and evaluated at observation noise σ , define the empirical radius–margin pass event*

$$\widehat{\mathcal{G}}_{\rho, \sigma} = \left\{ \widehat{Q}_{1-\alpha}(R_{\sigma}) \leq \widehat{r}_{\text{margin}} \right\} \cap \left\{ \widehat{\text{SMPR}}_{\rho, \sigma} \geq 1 - \widehat{\gamma} \right\}.$$

The matched-perturbation diagnostic region is the set of checkpoint indices ρ for which $\widehat{\mathcal{G}}_{\rho, \sigma}$ holds. If planner-margin traces are unavailable, $\widehat{r}_{\text{margin}}$ must be reported as a task-grounded rollout-space proxy, not as a planner-margin certificate. If diagnostics are available only at the no-noise and recovered endpoints, the evidence is endpoint mechanism evidence rather than interval identification.

3.5 Sampled-pool tail motivation and Gaussian sensitivity

The fixed-candidate statements motivate the deterministic link from paired rollout agreement to candidate decisions. A once-sampled candidate pool adds a tail requirement: if a single candidate has $\Pr[D > \epsilon] \leq \delta$, then a pool of size K can still contain a bad clean/noisy rollout pair with probability up to $K\delta$. A flip can also occur when the clean top-1/top-2 margin is at most $2L_J\epsilon$. This is the reason ATR uses an upper-tail summary rather than a mean disagreement score, and it is also why ATR is not a closed-loop control guarantee. Appendix A gives the proof.

The Gaussian stressor also has a local sensitivity interpretation. Let $G_{\mathbf{a}}(z) = \Pi(F_{\theta}^{1:H}(z, \mathbf{a}))$ and let $\tilde{o} = o + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 1 (Local Gaussian ACPC sensitivity). *If E_{θ} and $G_{\mathbf{a}}$ are locally twice differentiable around o and $E_{\theta}(o)$ with bounded second-order remainders, then for small σ ,*

$$G_{\mathbf{a}}(E_{\theta}(o + \xi)) - G_{\mathbf{a}}(E_{\theta}(o)) = J_{G_{\mathbf{a}}}(E_{\theta}(o))J_E(o)\xi + R_{\xi},$$

where the remainder contributes higher-order terms, and

$$\mathbb{E}_{\xi} \left[\|G_{\mathbf{a}}(E_{\theta}(o + \xi)) - G_{\mathbf{a}}(E_{\theta}(o))\|_2^2 \right] = \sigma^2 \|J_{G_{\mathbf{a}}}(E_{\theta}(o))J_E(o)\|_F^2 + O(\sigma^3).$$

The same linearization gives the local tail interpretation used by ATR.

Corollary 1 (Local Gaussian ACPC radius quantile). *Under the local linearization in Proposition 1, let*

$$A_{h,\mathbf{a}} = J_{G_{\mathbf{a}}}(E_{\theta}(o))J_E(o),$$

and let $\lambda_1, \dots, \lambda_r$ be the nonzero eigenvalues of $A_{h,\mathbf{a}}A_{h,\mathbf{a}}^{\top}$. For isotropic pixel noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$,

$$R_{\sigma}^2(h, \mathbf{a}) \approx \sigma^2 \sum_{i=1}^r \lambda_i \chi_i^2,$$

and therefore

$$r_{1-\alpha}(\sigma; h, \mathbf{a}) = Q_{1-\alpha}(R_{\sigma}) \approx \sigma \sqrt{Q_{1-\alpha} \left(\sum_i \lambda_i \chi_i^2 \right)}.$$

This corollary connects the empirical ATR quantile to the singular spectrum of the action-conditioned encoder–predictor Jacobian. ATR decreases when the composed map contracts along nuisance-noise directions, not necessarily when raw encoder distance alone is minimized. A useful diagnostic decomposition is

$$S_{\text{comp}}(h, \mathbf{a}) = \|J_{G_{\mathbf{a}}}(E_{\theta}(o))J_E(o)\|_F^2 = \text{tr}(J_E^{\top} J_{G_{\mathbf{a}}}^{\top} J_{G_{\mathbf{a}}} J_E),$$

with encoder-side and rollout-side reference scales

$$S_E(o) = \|J_E(o)\|_F^2, \quad S_G(E_{\theta}(o), \mathbf{a}) = \|J_{G_{\mathbf{a}}}(E_{\theta}(o))\|_F^2.$$

Noise training can reduce the composed sensitivity through encoder-side repair, rollout-side contraction, or alignment repair: the encoder’s high-sensitivity pixel directions may no longer feed the rollout map’s high-gain latent directions. We use the normalized diagnostic ratio

$$\kappa(h, \mathbf{a}) = \frac{\|J_{G_{\mathbf{a}}} J_E\|_F^2}{\|J_{G_{\mathbf{a}}}\|_F^2 \|J_E\|_F^2 + \varepsilon}$$

only as an orientation for mechanism audits, not as an angle certificate.

Combining the local radius quantile with the fixed-pool margin condition gives the diagnostic noise scale

$$\sigma^*(h, \mathbf{a}, \mathcal{A}) = \frac{\Delta_{\mathcal{A}}}{2L_J \sqrt{Q_{1-\alpha}(\sum_i \lambda_i \chi_i^2)}}.$$

This is a local diagnostic approximation, not a global robustness theorem. When L_J or the weighted-chi-square tail is unavailable, the empirical cost-drift slope $\hat{s}_C = \hat{Q}_q(|C_h - C_{\tilde{h}}|)/\sigma$ gives only a proxy tolerance $\hat{\sigma}_{\text{cost}}^* = \hat{Q}_{0.50}(\Delta_{\mathcal{A}})/(2\hat{s}_C)$.

Thus Gaussian ACPC measures action-conditioned encoder–predictor sensitivity, not encoder invariance alone. Noise training can improve robustness through both factors: it can reduce same-state encoder neighborhood crossing, and it can reduce the rollout sensitivity applied to any remaining encoder shift. Small encoder shifts can still be amplified by J_{G_a} , while large nuisance shifts can become harmless after rollout if the product $J_{G_a}J_E$ is small in those directions. The selective target is therefore asymmetric: the product should be small along nuisance perturbation directions, while projected rollouts for action-relevant state differences remain separated. The task-grounded near-boundary proxy margin pass event used later is

$$M = \mathbf{1}[d_{\text{proxy diff}} > d_{\text{same state noise}} + \delta_m],$$

with $\delta_m = 0$ in the reported table. It is a finite state-proxy selective-consistency check, not a full semantic-label coverage guarantee.

The theoretical statements in this section are intended as diagnostic orientation rather than as closed-loop control guarantees. They identify the failure modes that a paired clean/noisy diagnostic should measure: rollout-disagreement tails, clean candidate-margin caveats, and collapse of task-grounded separations. They do not assert that a low empirical ATR is a uniform bound for adaptive planning, repeated replanning, or environment-feedback stability.

Collapse caveat. Low same-state ACPC alone is insufficient. A constant encoder–predictor can make every clean and perturbed rollout identical while also mapping action-distinct states to identical projected futures. This is why SMPR is part of the reported diagnostic: ATR measures predictive stability under nuisance perturbation, and SMPR checks that task-grounded separations have not been erased.

4 Study protocol

The protocol separates behavior endpoints from diagnostics. Closed-loop success under Gaussian evaluation noise establishes the behavioral phenomenon; ATR and SMPR are then computed on fixed checkpoints to localize whether recovered endpoints also satisfy the selective ACPC pattern. Training seed, checkpoint epoch, evaluation seeds, and trajectory counts are kept fixed across the main comparisons.

4.1 Models and noise intervention

LeWM [19] is an end-to-end JEPA world model trained with a latent prediction loss and SIGReg regularization for latent-space model-predictive control. PushT (2D pushing), TwoRoom (2D navigation), Reacher (2D arm), and Cube (3D manipulation) form a heterogeneous stress panel spanning different nuisance/discriminability regimes. We add per-frame Gaussian noise to the input sequence as a full-sequence augmentation: each frame is independently noised with probability $p = 1.0$, and the standard deviation is drawn from Uniform(0, σ_{max}). This is a full-sequence input-side augmentation. We sweep $\sigma_{\text{max}} \in \{0.01, \dots, 0.08\}$ and evaluate observation-only Gaussian noise with the goal image unperturbed as the primary robustness endpoint.

4.2 Evaluation and diagnostics

The canonical LeWM development grid contains 36 epoch-10 checkpoints for training seed 3072: four tasks and, for each task, a no-noise baseline plus eight noise-trained checkpoints. Each checkpoint is evaluated with three evaluation seeds (42/43/44) and 100 trajectories per seed; canonical tables report the population mean $\pm$ population std across those three evaluation seeds. A completed Gaussian training-seed replication adds independent LeWM training seeds 3073 and 3074 under the same nine-point Gaussian sweep and evaluation protocol. Together, seed 3072 plus replication seeds 3073/3074 provide three training seeds for the main Gaussian closed-loop endpoint in Figure 1.

The main results follow a two-part evidence chain. Closed-loop figures and tables report training-seed mean/std for Gaussian robustness. ATR measures whether same-state clean/noisy rollout-disagreement tails contract at recovered endpoints, and SMPR checks whether task-grounded near-boundary separations survive that contraction. Unseen-stressor scores are auxiliary scope checks rather than additional core diagnostics.

5 Experiments

The experiments use additive Gaussian pixel noise as the primary controlled stressor for a representation-diagnostic question. We first show that the stressor changes closed-loop behavior. We then ask where recovery appears in the model, prioritizing cross-task ACPC/paired-rollout evidence and keeping task-specific representation measurements only as discriminability checks. The study is a matched-stressor trace through the encoder–predictor–planner chain; checkpoint ranking, external method comparison, and broad perturbation-family transfer are outside its scope.

All results use the protocol in Section 4: LeWM-base denotes the no-noise checkpoint, and LeWM+noise denotes the eight-level $\sigma_{\max}$ sweep.

5.1 JEPA control fragility under Gaussian observation noise

Table 1 reports LeWM-base success rates under observation-only Gaussian evaluation noise. Each training-seed value is first averaged over the three evaluation seeds; the table reports mean $\pm$ population std across the three training seeds. The goal image is kept clean in every column.

Table 1: LeWM-base under observation-only Gaussian evaluation noise. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed. The goal image is kept clean.

Task	eval $\sigma = 0$	eval $\sigma = 0.03$	eval $\sigma = 0.05$	eval $\sigma = 0.08$
TwoRoom	94.67 $\pm$ 0.72	92.56 $\pm$ 1.23	81.33 $\pm$ 6.77	68.78 $\pm$ 2.45
PushT	81.78 $\pm$ 4.89	53.56 $\pm$ 12.91	19.22 $\pm$ 9.12	7.22 $\pm$ 5.81
Reacher	59.22 $\pm$ 1.03	42.56 $\pm$ 2.99	33.56 $\pm$ 1.77	18.22 $\pm$ 0.42
Cube	65.22 $\pm$ 1.10	63.00 $\pm$ 1.19	46.67 $\pm$ 1.96	43.11 $\pm$ 3.27

LeWM-base is strong on unperturbed images, but observation noise alone produces large closed-loop losses at $\sigma = 0.08$: PushT loses 74.56 ± 5.55 points, Reacher 41.00 ± 1.44 points, TwoRoom 25.89 ± 2.38 points, and Cube 22.11 ± 2.78 points between the first and last columns. The progression across noise levels is smooth enough to show that the endpoint is not an isolated artifact of one severity setting. PushT degrades most sharply, consistent with contact-heavy continuous control being sensitive to visual precision.

5.2 Noise augmentation supplies recovered checkpoints for diagnosis

Figure 1 shows the full LeWM Gaussian sweep aggregated across training seeds 3072/3073/3074. Each plotted point first averages evaluation seeds 42/43/44 within a training seed and then reports mean and population std across the three training seeds. The main qualitative pattern is a broad, task-dependent recovery band rather than a stable universal checkpoint ranking.

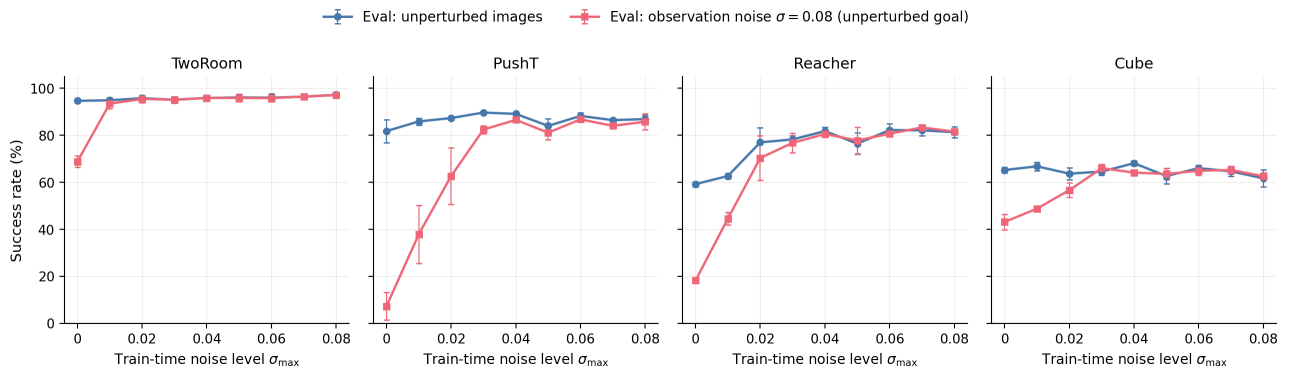


Figure 1: Three-training-seed LeWM Gaussian sweep across four tasks. Each point averages evaluation seeds 42/43/44 within each training seed and then reports mean across training seeds 3072/3073/3074; error bars show population std across training seeds. Blue circles: unperturbed evaluation; red squares: observation-only Gaussian noise $\sigma = 0.08$ with an unperturbed goal. The curves document broad task-dependent recovery bands rather than checkpoint rankings.

Because Figure 1 already aggregates the full sweep across three training seeds, we do not report a separate endpoint leaderboard in the main text. The fixed $\sigma_{\max} = 0.08$ endpoint substantially improves the observation-noise endpoint on TwoRoom, PushT, and Reacher, with a weaker positive Cube signal. Exact all-seed sweep values are reported in Appendix C for reproducibility.

A three-seed blur/resize score check later in this section serves as an auxiliary severe-stressor check outside the matched-Gaussian endpoint. The next subsection asks whether the recovered Gaussian endpoints also move in the ATR/SMPR diagnostics.

5.3 ATR/SMPR selective-ACPC diagnostics

For the main diagnostic, ATR is the 90th percentile of same-state clean/noisy ACPC-H rollout distance normalized by the clean transition scale. SMPR is the task-grounded near-boundary selective-margin pass rate at margin zero. Both are computed without retraining on the fixed no-noise and $\sigma_{\max} = 0.08$ checkpoints for training seeds 3072/3073/3074. Table 2 reports population mean/std across training seeds.

Table 2: ATR/SMPR selective-ACPC diagnostics over training seeds 3072/3073/3074. ATR is the q90 normalized same-state clean/noisy action-conditioned rollout disagreement; lower is better. SMPR is the task-grounded near-boundary selective-margin pass rate; higher is better.

Task	ATR base	ATR std0.08	SMPR base	SMPR std0.08
TwoRoom	1.509 ± 0.010	0.111 ± 0.016	0.34 ± 0.05	0.99 ± 0.01
PushT	3.580 ± 0.260	0.247 ± 0.017	0.44 ± 0.17	1.00 ± 0.00
Reacher	2.628 ± 0.048	0.082 ± 0.005	0.73 ± 0.03	1.00 ± 0.00
Cube	2.320 ± 0.045	0.100 ± 0.007	0.45 ± 0.03	1.00 ± 0.00

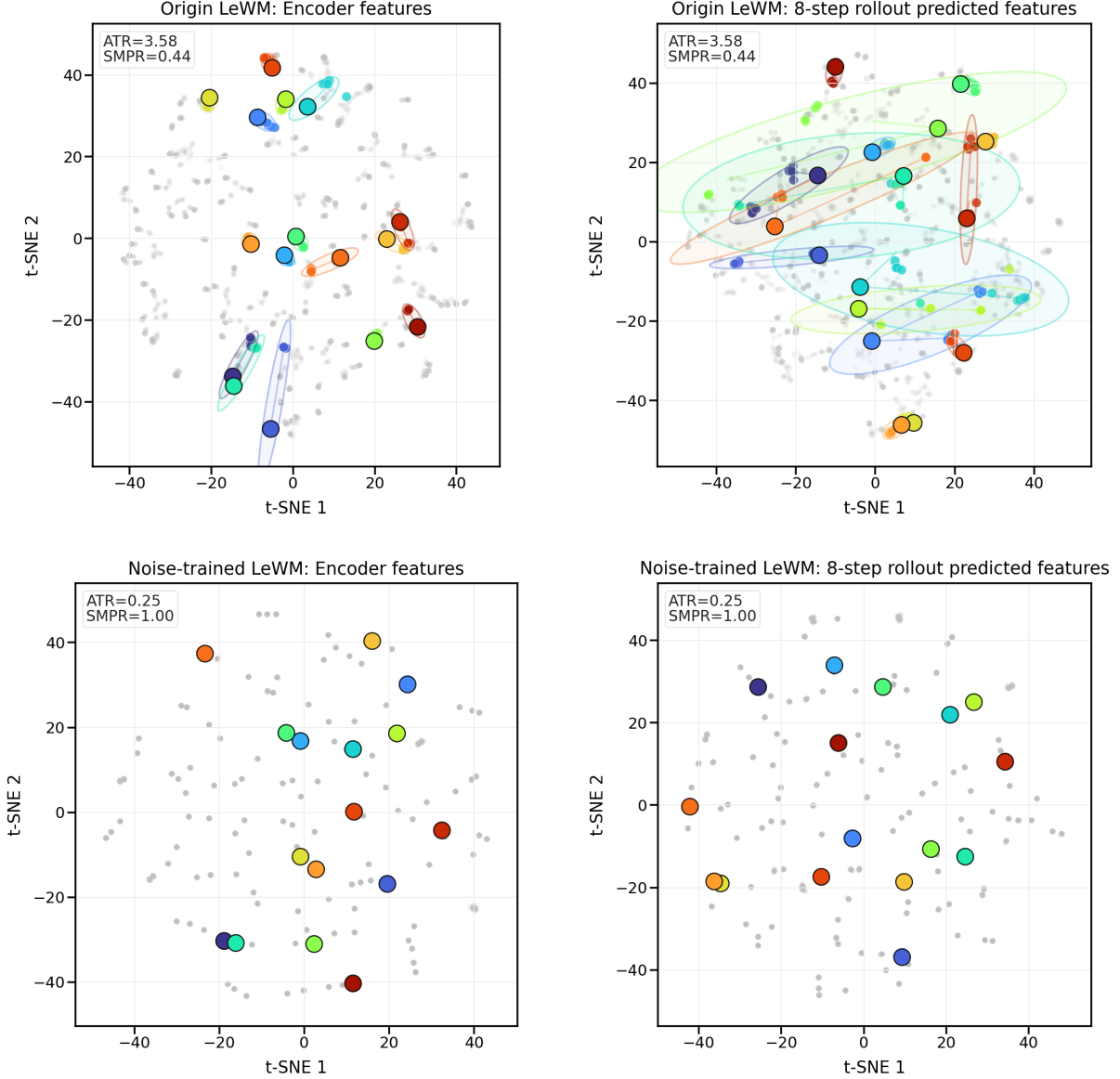


Figure 2: Qualitative PushT ACPC neighborhood t-SNE visualization for the Origin LeWM versus $\sigma_{\max} = 0.08$ noise-trained contrast. The four panels show encoder features and 8-step rollout predicted features for repeated same-state perturbation views; colored clusters mark fixed random anchors. The t-SNE projection is qualitative. ATR and SMPR are the quantitative diagnostics and are computed in the original projected-rollout diagnostic space.

The two diagnostics move in the intended selective direction on all four tasks: recovered endpoints have much smaller same-state predictive tails and much higher task-grounded margin pass rates. Figure 2 provides qualitative visual intuition on PushT: repeated noisy views of the same state become tighter after noise training in both encoder features and 8-step rollout predicted-feature projections. We interpret this as encoder basin repair plus further predictive contraction of residual nuisance variation, while keeping the quantitative claim on ATR, SMPR, and closed-loop behavior rather than distances in the t-SNE plane. Cube remains the useful boundary case: ATR and SMPR improve strongly, but the closed-loop Gaussian recovery in Figure 1 is weaker than PushT or Reacher, so closed-loop evaluation remains the behavioral authority.

SMPR uses programmatic task-state proxy labels, not oracle semantic annotations: TwoRoom uses doorway/room-side labels, PushT uses T-block pose cells, Reacher uses target/end-effector relation bins, and Cube uses cube-pose/goal-relation cells. Each anchor selects the closest label-crossing neighbor inside the closest

35% state-distance neighborhood and passes if that clean different-state rollout distance exceeds the same-state noisy radius. The no-noise baselines are weak under this stricter audit, while the high-noise endpoint reaches near-one pass rate; stronger hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels remain future validation.

5.4 Full-sweep diagnostic dynamics and held-out validation

Endpoint diagnostics are supplemented by a training-free full-sweep audit over all $4 \times 3 \times 9 = 108$ task-training-seed-checkpoint rows. Figure 3 overlays the closed-loop observation-noise score with normalized ATR tail, SMPR failure ($1 - \text{SMPR}$), and retained fixed-pool top-1 disagreement. The recovered portions of the Gaussian sweep consistently occupy lower normalized ATR and higher SMPR than fragile rows. We also recompute the fixed 65-candidate pool from checkpoints for all 108 rows, keeping sample-level maximum cost drift and observed top-1 flips. In Table 9, recovered rows have higher sufficient-event pass rates and much lower fixed-pool top-1 flip rates, while strict q10/q95 gaps remain negative; Table 10 adds Wilson intervals for the sampled event rates. This strengthens the mechanism audit without turning it into a calibrated probability certificate.

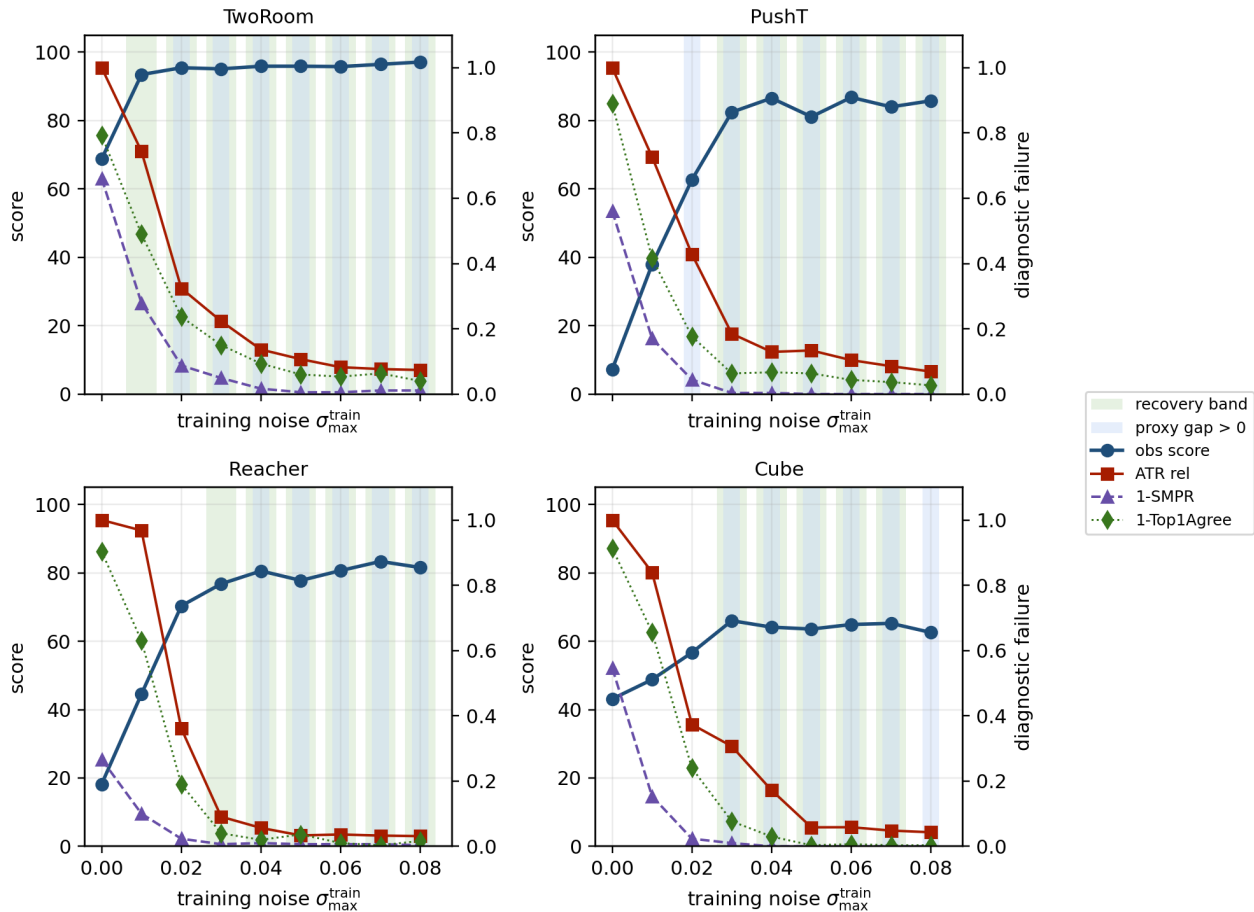


Figure 3: Full-sweep diagnostic dynamics across the Gaussian training grid. Each task panel averages training seeds 3072/3073/3074 at the same $\sigma_{\max}$ value. Green bands mark behavioral recovery-band points; blue bands mark positive retained fixed-pool proxy gaps. The right-axis curves show normalized ATR tail and SMPR failure ($1 - \text{SMPR}$), plus fixed-pool top-1 disagreement; lower is better for all three diagnostic-failure curves.

We also freeze simple diagnostic gates on calibration units before evaluating held-out units. In leave-one-training-seed-out validation, gates calibrated on two seeds and evaluated on the held-out seed have mean absolute recovery-onset error 0.007 in $\sigma_{\max}$ units, maximum error 0.020, precision 0.90, and recall 0.91. In leave-one-task-out validation, the corresponding values are 0.008, 0.020, 0.90, and 0.88. These held-out audits support ATR/SMPR as matched-Gaussian recovered-vs-fragile diagnostics, not as universal checkpoint rankers;

task-dependent boundary errors remain visible.

Table 3: Held-out diagnostic-region validation. Gates are selected on calibration seeds or tasks and evaluated on held-out sweeps. Start error is in $\sigma_{\max}^{\text{train}}$ units; positive means the diagnostic starts later than the behavioral recovery band.

Split	blocks	mean start err	max start err	precision	recall
seed held-out	12	0.007	0.020	0.90	0.91
task held-out	12	0.008	0.020	0.90	0.88

Threshold sensitivity is reported in Appendix B.2. Across recovery fractions 0.70/0.80/0.90 and clean-score tolerances of 3/5/10 points, all four tasks keep the expected recovered-vs-fragile direction: recovered rows have lower median normalized ATR and higher median SMPR than fragile rows. Table 12 keeps the interpretation joint: ATR is the radius term, while SMPR and fixed-pool top-1 flip are guard-side checks and are not standalone robustness metrics. ATR q80/q95 and positive-margin SMPR variants remain unavailable in the retained full-sweep ATR/SMPR artifact rather than inferred.

5.5 Diagnostic validation of the radius–margin certificate

The radius–margin theory predicts that a checkpoint should remain diagnostically stable under the matched stressor when the high-tail same-state predictive radius is contained within the relevant planner or task-discriminability margin. We therefore add a training-free full-sweep validation using existing three-training-seed evaluation and diagnostic summaries. The behavioral recovery band is defined only from closed-loop observation-noise scores: a checkpoint is marked as inside the recovery band if its observation-only $\sigma = 0.08$ success is at least 80% of the way from the no-noise baseline to the maximum grid value for that task, while clean success stays within 5 points of the no-noise clean score. The diagnostic proxy is computed independently from fixed-pool diagnostic summaries as

$$\hat{\Gamma}_{\rho,0.08} = \hat{Q}_{0.50}(\Delta_{\mathcal{A}}) - 2\hat{Q}_{0.90}(|C_h - C_{\tilde{h}}|).$$

A positive proxy gap is the available recorded-summary counterpart of the fixed-pool margin condition. The retained fixed-pool summaries store median clean candidate margin and q90 paired cost drift; they do not store the lower-tail margin and upper-tail drift summaries needed to instantiate a calibrated q10/q95 certificate. We therefore call this gap a mechanism proxy, not a planner-margin certificate. Thus this audit instantiates the theorem only at the proxy level supported by the recorded summaries: K is observed from the fixed candidate pool, while the single-candidate tail and failure probabilities are not calibrated. ATR-q90, the cost-margin proxy, and SMPR are empirical counterparts of the radius, planning-margin, and discriminability terms. Because the Gaussian sweep uses equal $\sigma_{\max}$ increments while the tasks have unequal visual and dynamical sensitivity, aggregate grid-point F1 or interval IoU is not used as the primary validation criterion. Appendix B.1 instead reports boundary-aware recovery/proxy alignment and the fixed-pool top-1 agreement audit used to interpret this diagnostic. The new sample-level full-sweep recomputation checks the sufficient event over the fixed 65-candidate pool for all 108 rows: across all tasks, median cert-pass rises from 0.06 on fragile rows to 0.61 on recovered rows, and median top-1 flip falls from 0.54 to 0.04. The strict q10/q95 gaps remain negative, so this strengthens the fixed-pool mechanism audit without turning it into a calibrated certificate.

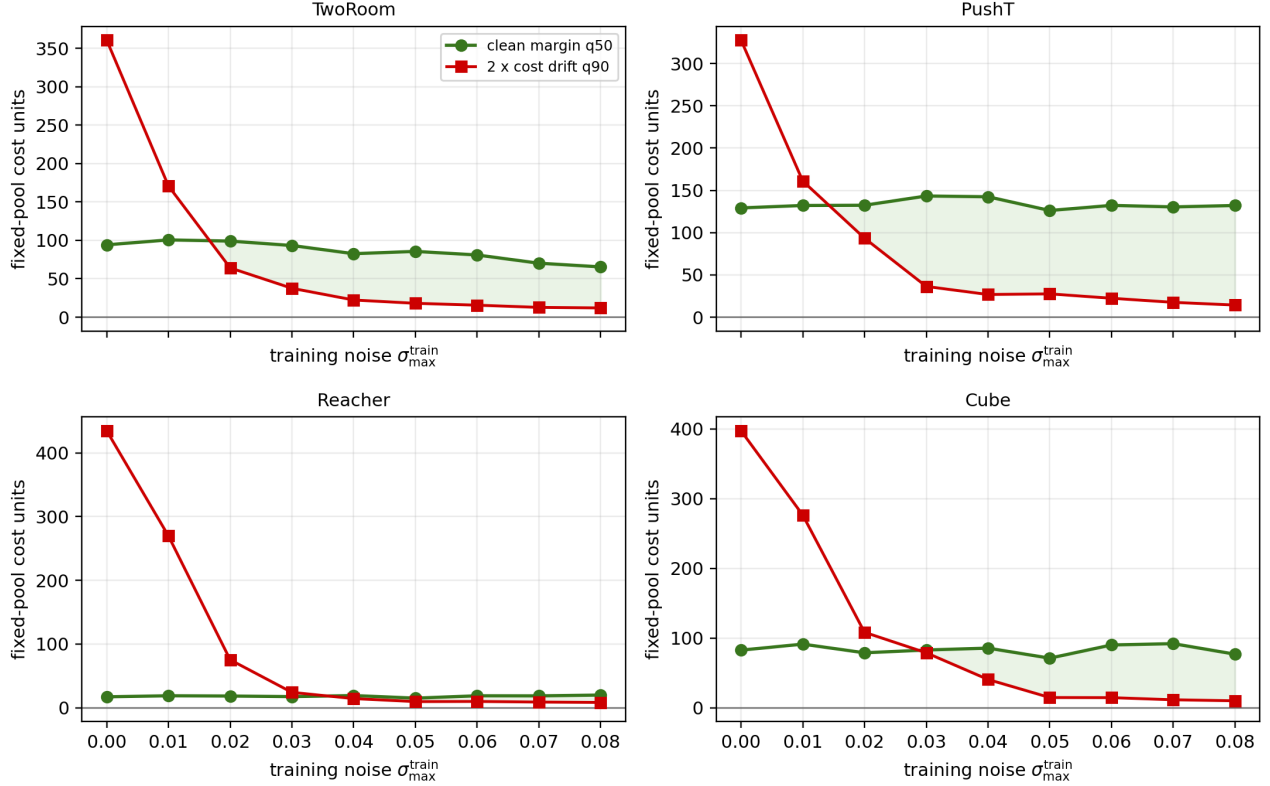


Figure 4: Fixed-pool radius–margin overlap proxy across the full Gaussian training sweep. Curves use the retained fixed-pool summaries: q50 clean candidate margin and twice the q90 paired cost drift. Positive separation supports the radius–margin mechanism at the recorded-summary level. A numerical q10/q95 planner–margin certificate would require recomputing lower-tail margin and upper-tail drift summaries from raw fixed-pool candidate traces or checkpoints, and the sampled pool is fixed rather than adaptive CEM.

The retained fixed-pool overlap proxy aligns with the broad recovery band but does not exactly reproduce it. TwoRoom and PushT cross the proxy gap within one grid step of the recovery-band onset, Cube aligns on the discrete grid, and Reacher is conservative by two grid steps. This is the expected level of evidence for a diagnostic certificate: it explains the radius–margin mechanism behind the observed recovery band without replacing closed-loop evaluation. The Reacher mismatch is informative rather than hidden: fixed-pool top-1 agreement already rises at the behavioral recovery onset, while the retained-summary cost–margin proxy becomes positive later. To keep the fixed-pool cost–margin proxy separate from a checkpoint-selection claim, we do not turn this proxy into a radius–margin ATR/SMPR gate table. The full-sweep ATR/SMPR held-out validation in Section 5.4 handles recovered-vs-fragile diagnostic separation; this subsection isolates the retained fixed-pool cost–margin mechanism and its limits.

5.6 Gaussian sensitivity audits

To check the local Gaussian sensitivity interpretation without claiming a global guarantee, we run two checkpoint-local audits on the base, recovery-onset, and endpoint checkpoints. First, a finite-difference audit perturbs the same observation histories with small Gaussian noise levels and estimates $\mathbb{E}[R_{\sigma}^2]/\sigma^2$ from the action-conditioned rollout radius. Table 13 reports a consistent endpoint/base reduction in this local slope, from 0.002 to 0.008 of the no-noise-checkpoint value across tasks, alongside the ATR contraction. Second, Table 14 estimates exact-autograd JVP/Hutchinson Frobenius traces for the encoder map, the rollout map, and their composition. The composed trace falls sharply at the endpoint on all tasks (endpoint/base ratios 0.003, 0.010, 0.006, and 0.028 for TwoRoom, PushT, Reacher, and Cube). The decomposition attributes this mainly to encoder-side sensitivity reduction; the rollout-side trace is not uniformly smaller, so we do not claim a standalone predictor-Jacobian repair. These audits support the composed encoder–predictor sensitivity explanation while remaining local mechanism checks rather than closed-loop robustness proofs.

5.7 Bounded unseen-stressor check

The main evidence remains the matched Gaussian-noise endpoint. After freezing the Gaussian grid, we ran strongest-severity unseen-stressor score checks to test whether the fixed $\sigma_{\max} = 0.08$ endpoint transfers outside the training perturbation family. Table 4 reports the non-Gaussian behavior together with stressor-specific ATR/SMPR diagnostics recomputed under the same blur or resize perturbation. TwoRoom and Reacher improve under strongest blur, whereas PushT and Cube do not show a clear resize gain at the scale of training-seed and evaluation variance. We treat this as bounded behavior outside the matched Gaussian setting.

Table 4: Bounded unseen-stressor score check with stressor-specific ATR/SMPR diagnostics. Score values are mean $\pm$ population std across LeWM training seeds 3072/3073/3074; each seed point averages evaluation seeds 42/43/44 from strongest-severity unseen runs. The ATR and SMPR columns report raw no-noise $\rightarrow$ noise-trained diagnostic values under the row stressor; lower ATR and higher SMPR are better.

Task	evaluation stressor	no-noise score	noise-trained score	ATR	SMPR
TwoRoom	blur $k = 15$	47.67 ± 5.44	90.78 ± 5.38	$1.61 \rightarrow 1.24$	$0.16 \rightarrow 0.77$
Reacher	blur $k = 15$	22.00 ± 3.78	71.22 ± 1.10	$2.81 \rightarrow 0.54$	$0.60 \rightarrow 0.98$
PushT	resize 0.25	63.44 ± 14.05	66.33 ± 8.38	$1.77 \rightarrow 1.53$	$0.93 \rightarrow 0.96$
Cube	resize 0.25	57.00 ± 1.96	56.11 ± 0.57	$1.35 \rightarrow 1.59$	$0.98 \rightarrow 0.95$

The raw unseen diagnostics co-move with behavior at this severe-stressor scale. The blur rows combine large behavior gains with lower ATR and higher SMPR after noise training; PushT resize has only small score and diagnostic movement; Cube resize is neutral to slightly negative on behavior and also moves in the wrong diagnostic direction. These rows therefore support only a bounded severe-stressor association rather than a general perturbation-transfer claim.

6 Discussion and limitations

What the evidence supports. This is a controlled diagnostic study, not a method paper. Closed-loop evaluation establishes the behavior; ATR checks whether same-state noisy views have small action-conditioned predictive tails; SMPR checks whether task-grounded near-boundary distinctions survive that contraction. Full-sweep and held-out audits support recovered-vs-fragile diagnostic separation under the matched Gaussian stressor. The radius-margin audit adds fixed-pool cost-margin and sample-level event-rate evidence over the Gaussian grid, and the finite-difference and exact-JVP/Hutchinson audits check the local Gaussian sensitivity mechanism; mismatches are reported as diagnostic limits rather than hidden by the endpoint result. The primary statistic is the full three-training-seed Gaussian replication, with evaluation-seed variance kept as the within-seed measurement scale. These standard deviations are descriptive over three training seeds, not asymptotic significance claims.

Scope. Gaussian pixel noise remains the training/evaluation axis. The blur/resize rows are bounded behavior outside the matched Gaussian setting, not a transfer theorem. The radius-margin certificate is fixed-pool and matched-perturbation only: it does not cover adaptive CEM resampling, repeated replanning, or environment-feedback trajectory guarantees. The Gaussian sensitivity audits are local: finite-difference slopes and exact-JVP/Hutchinson trace estimates do not provide a global robustness or closed-loop guarantee. The empirical fixed-stressor diagnostic region is an audit view over matched checkpoints rather than a closed-loop guarantee. ATR/SMPR are not a checkpoint-ranker for $\sigma_{\max}$ and do not replace closed-loop evaluation inside a broad recovery band; SMPR is only a guard-side component of the joint diagnostic. We do not claim superiority over DrQ-style augmentation, TD-MPC2, DreamerV3, or robust MPC methods; the contribution is a no-retraining diagnostic for JEPa latent world-model checkpoints under a controlled Gaussian stressor.

What remains. The results argue against a too-strong reading of latent prediction: future-latent prediction alone does not guarantee closed-loop visual robustness. The joint guard-side audit checks that the low-ATR region also preserves SMPR and fixed-pool candidate stability, but stronger semantic discriminability evidence would still require hand-labeled or simulator-derived contact, topology, action-value, or cost-to-go labels, especially for PushT and Cube. Future methods can turn ATR/SMPR into objectives, but this paper keeps that as a separate method-paper path after diagnostic validity is established.

7 Conclusion

This paper frames Gaussian-noise robustness in JEPA latent world-model control as a selective diagnostic problem. Closed-loop behavior establishes the phenomenon; ATR measures same-state predictive-stability tails; SMPR checks task-grounded anti-collapse margins. On fixed LeWM checkpoints, input-side noise training recovers broad matched-Gaussian recovery bands across three training seeds, and ATR/SMPR move in the expected direction at recovered endpoints.

The theory is deliberately diagnostic: sampled-pool ACPC exposes why tail risk matters and why clean margins prevent closed-loop guarantees, while local Gaussian linearization connects the measured tail to action-conditioned encoder–predictor sensitivity. The resulting view is a radius–margin diagnostic theory for fixed-checkpoint Gaussian robustness: ATR estimates the perturbation-tube radius tail, SMPR estimates margin preservation, and their conjunction defines the empirical fixed-stressor diagnostic region studied here. The broader implication is not solved generalization; it is a concrete audit target for future methods: stable action-conditioned predictive dynamics under nuisance variation, while preserving distinctions needed for planning.

References

- [1] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023.
- [2] Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zhohus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-jepa 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025.
- [3] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mido Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024.
- [4] Yuxin Chen, Jianglan Wei, Chenfeng Xu, Boyi Li, Masayoshi Tomizuka, Andrea Bajcsy, and Ran Tian. Reimagination with test-time observation interventions: Distractor-robust world model predictions for visual model predictive control, 2025.
- [5] Tom Dupuis, Jaonary Rabarisoa, Quoc-Cuong Pham, and David Filliat. VIBR: Learning view-invariant value functions for robust visual control. In *Proceedings of The 2nd Conference on Lifelong Learning Agents*, volume 232 of *Proceedings of Machine Learning Research*, pages 658–682. PMLR, 2023.
- [6] T. W. Epps and Lawrence B. Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.
- [7] Carles Gelada, Saurabh Kumar, Jacob Buckman, Ofir Nachum, and Marc G. Bellemare. DeepMDP: Learning continuous latent space models for representation learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2170–2179. PMLR, 2019.
- [8] Hafez Ghaemi, Eilif Benjamin Muller, and Shahab Bakhtiari. seq-jepa: Autoregressive predictive learning of invariant-equivariant world models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- [9] Christopher Grimm, Andre Barreto, Satinder Singh, and David Silver. The value equivalence principle for model-based reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

- [10] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640(8059):647–653, 2025.
- [11] Nicklas Hansen, Hao Su, and Xiaolong Wang. Td-mpc2: Scalable, robust world models for continuous control. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- [12] Nicklas Hansen and Xiaolong Wang. Generalization in reinforcement learning by soft data augmentation. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 13611–13617, 2021.
- [13] Yongchao Huang. Vjepa: Variational joint embedding predictive architectures as probabilistic world models. *arXiv preprint arXiv:2601.14354*, 2026.
- [14] David Klindt, Yann LeCun, and Randall Balestriero. When does lejepa learn a world model?, 2026.
- [15] Ilya Kostrikov, Denis Yarats, and Rob Fergus. Image augmentation is all you need: Regularizing deep reinforcement learning from pixels. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [16] Yann LeCun. A path towards autonomous machine intelligence. OpenReview preprint, 2022.
- [17] Etai Littwin, Omid Saremi, Madhu Advani, Vimal Thilak, Preetum Nakkiran, Chen Huang, and Joshua Susskind. How JEPA avoids noisy features: The implicit bias of deep linear self distillation networks, 2024.
- [18] Lucas Maes, Quentin Le Lidec, Luiz Facury, Nassim Massaudi, Ayush Chaurasia, Francesco Capuano, Richard Gao, Taj Gillin, Dan Haramati, Damien Scieur, Yann LeCun, and Randall Balestriero. stable-worldmodel: A platform for reproducible world modeling research and evaluation, 2026.
- [19] Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworldmodel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.
- [20] Lorenzo Mur-Labadia, Matthew Muckley, Amir Bar, Mido Assran, Koustuv Sinha, Mike Rabbat, Yann LeCun, Nicolas Ballas, and Adrien Bardes. V-jepa 2.1: Unlocking dense features in video self-supervised learning. *arXiv preprint arXiv:2603.14482*, 2026.
- [21] Mingyu Park, Samyeul Noh, Hyun Myung, and Donghwan Lee. Zero-shot visual generalization in model-based reinforcement learning via latent consistency. OpenReview (ICLR 2026 submission), 2025.
- [22] Yuping Qiu, Rui Zhu, and Ying-cong Chen. Improving joint embedding predictive architecture with diffusion noise. *arXiv preprint arXiv:2507.15216*, 2025.
- [23] Ashwath Radhachandran, Vedrana Ivezić, Shreeram Athreya, Ronit Anilkumar, Corey W. Arnold, and William Speier. Us-jepa: A joint embedding predictive architecture for medical ultrasound. *arXiv preprint arXiv:2602.19322*, 2026.
- [24] Yutaka Shimizu and Masayoshi Tomizuka. Bisimulation metric for model predictive control. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025.
- [25] Vlad Sobal, Jyothir S V, Siddhartha Jalagam, Nicolas Carion, Kyunghyun Cho, and Yann LeCun. Joint embedding predictive architectures focus on slow features, 2022.
- [26] Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Stress-testing offline reward-free reinforcement learning: A case for planning with latent dynamics models. In *WRL@ICLR 2025*, 2025.
- [27] Alex Tamkin, Margalit Glasgow, Xiluo He, and Noah Goodman. Feature dropout: Revisiting the role of augmentations in contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [28] Leonardo F. Toso, Davit Shadunts, Yunyang Lu, Nihal Sharma, Donglin Zhan, Nam H. Nguyen, and James Anderson. Learning invariant visual representations for planning with joint-embedding predictive world models. *arXiv preprint arXiv:2602.18639*, 2026.

- [29] Hugues Van Assel, Mark Ibrahim, Tommaso Biancalani, Aviv Regev, and Randall Balestriero. Joint embedding vs reconstruction: Provable benefits of latent space prediction for self-supervised learning, 2025.
- [30] Claas A. Voelcker, Anastasiia Pedan, Arash Ahmadian, Romina Abachi, Igor Gilitschenski, and Amir-Massoud Farahmand. Calibrated value-aware model learning with probabilistic environment models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 61745–61768. PMLR, 2025.
- [31] Tongzhou Wang and Phillip Isola. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 9929–9939. PMLR, 2020.
- [32] Zijie Wang, Wei Zhang, Weiming Zhang, Fanqi Zhang, Xiao Tan, Yipeng Qin, and Guanbin Li. World models as group actions, 2026.
- [33] Denis Yarats, Rob Fergus, Alessandro Lazaric, and Lerrel Pinto. Mastering visual continuous control: Improved data-augmented reinforcement learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2022.
- [34] Amy Zhang, Rowan T. McAllister, Roberto Calandra, Yarin Gal, and Sergey Levine. Learning invariant representations for reinforcement learning without reconstruction. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
- [35] Junbo Zhang and Kaisheng Ma. Rethinking the augmentation module in contrastive learning: Learning hierarchical augmentation invariance with expanded views. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16650–16659, 2022.

A Proofs for ACPC Diagnostics

These proofs support the diagnostic use of ATR and SMPR; they do not provide closed-loop control guarantees.

Proposition 2 (ACPC controls candidate-cost drift). *Assume that J is locally L_J -Lipschitz on the clean/perturbed projected-rollout neighborhood evaluated by the fixed candidate set under metric d_H . If*

$$d_H(\Pi(F_\theta^{1:H}(E_\theta(h), \mathbf{a})), \Pi(F_\theta^{1:H}(E_\theta(\tilde{h}), \mathbf{a}))) \leq \epsilon,$$

then

$$|C_h(\mathbf{a}, g) - C_{\tilde{h}}(\mathbf{a}, g)| \leq L_J \epsilon.$$

Proof. The planner cost J is L_J -Lipschitz on the projected-rollout sequence, so the cost difference is bounded by L_J times the clean/perturbed projected-rollout distance. $\square$

Proposition 3 (Candidate top-1 stability). *Let $\mathcal{A} = \{\mathbf{a}^1, \dots, \mathbf{a}^K\}$ be the shared candidate set. Assume lower cost is preferred, and let $\mathbf{a}^{(1)}$ and $\mathbf{a}^{(2)}$ denote the lowest- and second-lowest-cost candidates on the clean branch. Suppose that for every candidate j ,*

$$|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq \eta.$$

If the clean branch has a strict top-1/top-2 margin

$$\Delta = C_h(\mathbf{a}^{(2)}, g) - C_h(\mathbf{a}^{(1)}, g) > 2\eta,$$

then the perturbed branch selects the same top-1 candidate.

Proof. For any candidate $\mathbf{a}^j \neq \mathbf{a}^{(1)}$,

$$C_{\tilde{h}}(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^{(1)}, g) \geq C_h(\mathbf{a}^j, g) - C_h(\mathbf{a}^{(1)}, g) - 2\eta \geq \Delta - 2\eta > 0.$$

Thus every non-top candidate remains more costly than the clean top-1 candidate on the perturbed branch. $\square$

Proof of Theorem 1.

Proof. Let $\mathcal{B} = \{\max_{1 \leq j \leq K} D_j \leq \epsilon\}$. Since each candidate is drawn from q and a single draw has tail probability at most α , the union bound gives

$$\Pr_{\mathcal{A}}(\mathcal{B}^c) \leq \sum_{j=1}^K \Pr[D_j > \epsilon] \leq K\alpha.$$

On $\mathcal{B}$, local Lipschitzness of the planner cost gives $|C_h(\mathbf{a}^j, g) - C_{\tilde{h}}(\mathbf{a}^j, g)| \leq L_J \epsilon$ for every candidate. By the top-1 stability proposition, the perturbed branch can choose a different top candidate on $\mathcal{B}$ only when $\Delta_{\mathcal{A}} \leq 2L_J \epsilon$. Therefore the top-1 flip event is contained in

$$\mathcal{B}^c \cup \{\Delta_{\mathcal{A}} \leq 2L_J \epsilon\},$$

which yields the fixed-pool instability bound without any independence assumption beyond the single-candidate tail statement. For the selective diagnostic statement, let $\mathcal{D} = \{M_{\text{diff}} \leq R_\sigma + \delta_m\}$. Selective failure is contained in the union of the fixed-pool planning-instability event and $\mathcal{D}$. Applying the union bound again and using $\Pr(\mathcal{D}) \leq \gamma$ gives the final bound. $\square$

Local Gaussian sensitivity. For small Gaussian pixel noise ξ , a first-order expansion gives $E_\theta(o + \xi) = E_\theta(o) + J_E(o)\xi + \mathcal{R}(\xi)$ with $\|\mathcal{R}(\xi)\| = O(\|\xi\|^2)$. Composing the encoder with the action-conditioned rollout map shows that same-state rollout disagreement measures local sensitivity of the encoder–predictor composition, not encoder invariance alone. ATR reports a high-quantile version of that paired rollout disagreement.

B Experimental Details

These details define the fixed evaluation and diagnostic protocol used by the main tables.

LeWM is evaluated on PushT, TwoRoom, Reacher, and Cube using the canonical three evaluation seeds 42/43/44 and 100 trajectories per seed. The main Gaussian robustness endpoint perturbs observation pixels with Gaussian noise at standard deviation 0.08 while keeping the goal image clean. Noise-trained checkpoints use full-sequence input-side Gaussian augmentation with $\sigma_{\max} \in \{0.01, \dots, 0.08\}$.

ATR is computed at each task-training-seed-checkpoint row as the 90th percentile same-state ACPC-H distance normalized by the clean transition scale. SMPR uses task-grounded near-boundary proxy labels at margin zero. The paper reports population mean/std across training seeds 3072/3073/3074.

B.1 Radius-margin parameter interpretation

Theorem 1 separates the quantities needed for a numerical fixed-pool certificate from the lower-sample diagnostic summaries reported in the experiments. In the fixed-pool diagnostic summaries used for Figures 3 and 4, the candidate count is $K = 65$ for every task-checkpoint row. The theorem’s single-candidate tail probability α is not fixed by the q90 reporting level used by ATR. A q90 summary descriptively leaves 10% of recorded same-state radii above the reported quantile, but that descriptive exceedance rate is not the α used in the theorem. If it were naively inserted into the union-bound term, the result would be $65 \times 0.1 = 6.5$, which is vacuous. A non-vacuous numerical candidate-flip bound would require a much smaller single-candidate tail probability or a different concentration argument. We therefore use q90 ATR and the retained q50/q90 cost-margin overlap as diagnostic evidence for the radius-margin mechanism, not as a calibrated probability guarantee.

Table 5: Fixed-pool radius-margin interpretation used in the diagnostic validation. The behavioral recovery band is defined from closed-loop observation-only $\sigma = 0.08$ scores; the proxy-positive range uses $\widehat{Q}_{0.50}(\Delta_{\mathcal{A}}) - 2\widehat{Q}_{0.90}(|C_h - C_{\bar{h}}|) > 0$ on the recorded fixed-pool summaries. These ranges explain the recovery/proxy bands in Figure 3 and the overlap proxy in Figure 4; they are not used to claim a calibrated flip-probability certificate.

Task	K	recovery band	proxy-positive range	interpretation
TwoRoom	65	0.01–0.08	0.02–0.08	proxy misses the first recovery-band checkpoint
PushT	65	0.03–0.08	0.02–0.08	proxy includes one earlier checkpoint
Reacher	65	0.02–0.08	0.04–0.08	proxy is conservative at low noise-training levels
Cube	65	0.03–0.08	0.03–0.08	proxy and recovery-band ranges align

The one-grid tolerance is $\tau_{\text{grid}} = 0.01$ in $\sigma_{\max}$ units. Start-boundary error is defined as $\rho_{\text{start}}^{\text{diag}} - \rho_{\text{start}}^{\text{beh}}$, so positive values mean the diagnostic proxy is later and negative values mean it fires earlier.

Table 6: Boundary-aware interpretation of the fixed-pool radius-margin proxy. Equal spacing in $\sigma_{\max}$ is not equal spacing in task sensitivity, so these rows are interpreted as task-conditioned boundary checks rather than grid-point classifier scores.

Task	recovery band	proxy range	start error	within τ_{grid} ?	interpretation
TwoRoom	0.01–0.08	0.02–0.08	+0.01	yes	proxy misses the first recovery-band checkpoint
PushT	0.03–0.08	0.02–0.08	−0.01	yes	proxy fires one grid step early
Reacher	0.02–0.08	0.04–0.08	+0.02	partial	fixed-pool cost-margin proxy is conservative
Cube	0.03–0.08	0.03–0.08	0.00	yes	proxy and recovery band align

The recorded fixed-pool flip rate is the clean/noisy top-1 disagreement rate under the same candidate set, so one minus this flip rate gives an empirical top-1 agreement audit that is closer to the theorem’s intermediate statement than closed-loop score. It still does not cover adaptive CEM resampling or repeated replanning.

Table 7: Fixed-pool top-1 agreement audit derived from the recorded fixed-pool flip rate. Rows show the no-noise baseline, the first closed-loop recovery-band checkpoint on the discrete sweep, and the matched $\sigma_{\max} = 0.08$ endpoint.

Task	checkpoint	cost drift q90	clean margin q50	proxy gap	Top1Agree
TwoRoom	base 0.00	180.33	94.14	-266.52	0.207
TwoRoom	onset 0.01	85.42	100.56	-70.28	0.510
TwoRoom	endpoint 0.08	6.00	65.46	53.46	0.960
PushT	base 0.00	163.82	129.21	-198.44	0.110
PushT	onset 0.03	18.16	143.29	106.96	0.937
PushT	endpoint 0.08	7.15	132.05	117.74	0.973
Reacher	base 0.00	217.30	17.29	-417.32	0.097
Reacher	onset 0.02	37.67	18.54	-56.80	0.810
Reacher	endpoint 0.08	4.24	20.08	11.59	0.983
Cube	base 0.00	198.70	82.76	-314.65	0.087
Cube	onset 0.03	39.48	82.95	3.99	0.923
Cube	endpoint 0.08	5.05	76.94	66.85	0.997

In this interpretation, β_{plan} and β_{disc} name empirical failure components rather than calibrated probabilities. The planning component is represented by the fixed-pool cost-margin proxy, top-1 agreement audit, and the sample-level event-rate recomputation; the discriminability component is represented by SMPR, equivalently its failure rate $1 - \text{SMPR}$. The retained-summary table reports q90 cost drift and q50 clean margin only, while Table 9 separately reports sample-level q10/q95 gaps and event rates. Because those strict q10/q95 gaps remain negative, we do not manufacture certificate-strength tail claims from the recomputation. The closed-loop recovery band explains the fixed-pool radius-margin validation rather than converting the separate ATR/SMPR diagnostic-region audit into a threshold-classifier claim.

B.2 Remediation audit tables

The held-out validation in Table 3 is supported by fixed-pool, sample-level, sensitivity, and threshold-sensitivity audits. Table 8 reports full-sweep top-1 agreement and proxy-gap rows selected from the retained fixed-pool summaries. Table 9 adds the full-sweep sample-level event-rate recomputation, and Table 10 adds Wilson intervals for the sample-level event rates; these intervals quantify estimation uncertainty, not calibrated theorem probabilities. Table 11 preserves the base/endpoint view. Table 12 reports ATR together with SMPR and fixed-pool top-1 flip to keep the guard-side interpretation joint rather than standalone. Table 13 reports the finite-difference local Gaussian sensitivity audit, and Table 14 reports the exact-JVP/Hutchinson local trace decomposition. Table 15 checks that recovered-vs-fragile ATR/SMPR direction is stable under reasonable behavioral-label thresholds; unavailable ATR/SMPR tail variants are recorded in the audit outputs rather than inferred.

Table 8: Retained-summary fixed-pool top-1 audit. Top1Agree is derived from the recorded fixed-pool flip rate; cert-pass rates are not reported because sample-level maximum cost-drift traces are not retained.

Task	row	$\sigma_{\max}^{\text{train}}$	Top1Agree	proxy gap	proxy pass rate
TwoRoom	base	0.00	0.207	-266.52	0.00
TwoRoom	proxy onset	0.02	0.763	34.93	1.00
TwoRoom	endpoint	0.08	0.960	53.46	1.00
PushT	base	0.00	0.110	-198.44	0.00
PushT	proxy onset	0.02	0.823	38.95	0.67
PushT	endpoint	0.08	0.973	117.74	1.00
Reacher	base	0.00	0.097	-417.32	0.00
Reacher	proxy onset	0.04	0.980	4.79	1.00
Reacher	endpoint	0.08	0.983	11.59	1.00
Cube	base	0.00	0.087	-314.65	0.00
Cube	proxy onset	0.03	0.923	3.99	0.67
Cube	endpoint	0.08	0.997	66.85	1.00

Table 9: Full-sweep sample-level fixed-pool event-rate audit. Rows split all task-seed- $\sigma_{\max}$ checkpoints by the closed-loop recovery-band label. Cert-pass is the fraction of sampled states satisfying the fixed-pool sufficient event; top-1 flip is the observed clean/noisy fixed-pool best-candidate disagreement. These event rates strengthen the mechanism audit but are not calibrated probability bounds.

Task	cert-pass fragile $\rightarrow$ recovered	top-1 flip fragile $\rightarrow$ recovered	q10/q95 gap fragile $\rightarrow$ recovered
TwoRoom	0.05 $\rightarrow$ 0.67	0.79 $\rightarrow$ 0.07	-1053.7 $\rightarrow$ -269.0
PushT	0.16 $\rightarrow$ 0.67	0.38 $\rightarrow$ 0.06	-797.6 $\rightarrow$ -193.5
Reacher	0.06 $\rightarrow$ 0.54	0.54 $\rightarrow$ 0.02	-873.0 $\rightarrow$ -29.3
Cube	0.08 $\rightarrow$ 0.78	0.53 $\rightarrow$ 0.01	-713.4 $\rightarrow$ -55.3
ALL	0.06 $\rightarrow$ 0.61	0.54 $\rightarrow$ 0.04	-847.7 $\rightarrow$ -136.6

Table 10: Wilson intervals for full-sweep sample-level fixed-pool event rates. Each cell reports rate [95% Wilson interval] over sampled audit anchors after splitting checkpoints by the closed-loop recovery-band label. These intervals quantify event-rate estimation uncertainty and are not calibrated theorem probability bounds.

Task	cert-pass fragile $\rightarrow$ recovered	top-1 flip fragile $\rightarrow$ recovered
TwoRoom	0.09 [0.06,0.12] $\rightarrow$ 0.61 [0.59,0.63]	0.72 [0.68,0.77] $\rightarrow$ 0.10 [0.08,0.11]
PushT	0.18 [0.16,0.21] $\rightarrow$ 0.65 [0.63,0.67]	0.52 [0.49,0.55] $\rightarrow$ 0.06 [0.05,0.07]
Reacher	0.11 [0.09,0.13] $\rightarrow$ 0.52 [0.50,0.54]	0.55 [0.51,0.58] $\rightarrow$ 0.03 [0.02,0.04]
Cube	0.32 [0.29,0.35] $\rightarrow$ 0.64 [0.62,0.67]	0.47 [0.44,0.50] $\rightarrow$ 0.04 [0.03,0.05]
ALL	0.20 [0.19,0.21] $\rightarrow$ 0.61 [0.59,0.62]	0.53 [0.52,0.55] $\rightarrow$ 0.06 [0.05,0.06]

Table 11: Sample-level fixed-pool endpoint audit. Cert-pass is the fraction of sampled states where the maximum paired candidate-cost drift over the fixed 65-candidate pool is below half the clean top-1/top-2 margin. Values are mean $\pm$ population std across training seeds 3072/3073/3074 with 100 sampled states per seed. Strict q10/q95 gaps remain negative, so these rows support a fixed-pool mechanism audit rather than a calibrated probability certificate.

Task	cert-pass base $\rightarrow$ std0.08	top-1 flip base $\rightarrow$ std0.08	q10/q95 gap base $\rightarrow$ std0.08
TwoRoom	0.05 $\pm$ 0.00 $\rightarrow$ 0.73 $\pm$ 0.03	0.80 $\pm$ 0.02 $\rightarrow$ 0.03 $\pm$ 0.00	-1084.6 $\pm$ 44.0 $\rightarrow$ -179.0 $\pm$ 44.2
PushT	0.01 $\pm$ 0.01 $\rightarrow$ 0.74 $\pm$ 0.04	0.92 $\pm$ 0.03 $\rightarrow$ 0.03 $\pm$ 0.02	-917.4 $\pm$ 44.3 $\rightarrow$ -126.2 $\pm$ 13.5
Reacher	0.00 $\pm$ 0.00 $\rightarrow$ 0.61 $\pm$ 0.01	0.88 $\pm$ 0.03 $\rightarrow$ 0.01 $\pm$ 0.00	-935.2 $\pm$ 32.1 $\rightarrow$ -24.7 $\pm$ 2.3
Cube	0.00 $\pm$ 0.00 $\rightarrow$ 0.83 $\pm$ 0.01	0.92 $\pm$ 0.03 $\rightarrow$ 0.00 $\pm$ 0.00	-1005.3 $\pm$ 13.9 $\rightarrow$ -31.3 $\pm$ 7.7

Table 12: Joint ATR plus guard-side validation across the Gaussian sweep. Rows are split by the closed-loop recovery-band label. ATR is the radius term; SMPR and fixed-pool top-1 flip are guard-side checks against task or action-candidate collapse and are not interpreted as standalone robustness metrics.

Task	ATR fragile $\rightarrow$ recovered	SMPR fragile $\rightarrow$ recovered	top-1 flip fragile $\rightarrow$ recovered
TwoRoom	1.00 $\rightarrow$ 0.12	0.36 $\rightarrow$ 0.98	0.79 $\rightarrow$ 0.07
PushT	0.72 $\rightarrow$ 0.10	0.85 $\rightarrow$ 1.00	0.38 $\rightarrow$ 0.06
Reacher	0.97 $\rightarrow$ 0.04	0.90 $\rightarrow$ 0.99	0.54 $\rightarrow$ 0.02
Cube	0.71 $\rightarrow$ 0.07	0.93 $\rightarrow$ 1.00	0.53 $\rightarrow$ 0.01
ALL	0.84 $\rightarrow$ 0.09	0.86 $\rightarrow$ 1.00	0.54 $\rightarrow$ 0.04

Table 13: Finite-difference Gaussian sensitivity audit. The slope is the mean small-noise rollout radius squared divided by σ^2 , summarized over the reported small-noise probes and training seeds. Lower endpoint/base ratios indicate reduced local composed encoder-predictor sensitivity. This is a local finite-difference proxy, not a global robustness guarantee.

Task	slope base $\rightarrow$ endpoint	endpoint/base	ATR base $\rightarrow$ endpoint
TwoRoom	4.16e+05 $\rightarrow$ 1.50e+03	0.004	1.51 $\rightarrow$ 0.11
PushT	1.87e+05 $\rightarrow$ 1.48e+03	0.008	3.58 $\rightarrow$ 0.25
Reacher	9.22e+04 $\rightarrow$ 1.64e+02	0.002	2.63 $\rightarrow$ 0.08
Cube	5.23e+04 $\rightarrow$ 4.40e+02	0.008	2.32 $\rightarrow$ 0.10

Table 14: Exact-JVP/Hutchinson local sensitivity decomposition. Values are endpoint/base ratios of Hutchinson trace estimates, using 16 sampled sequences and 8 Rademacher probes per checkpoint, then taking medians over training seeds. Columns report encoder trace per history-pixel dimension, rollout trace per latent-history dimension, composed encoder-rollout trace per history-pixel dimension, and the alignment coefficient $\text{tr}(J_E^\top J_G^\top J_G J_E)/(\text{tr}(J_E^\top J_E)\text{tr}(J_G^\top J_G)/d_z)$. The audit uses exact autograd JVPs with math/eager attention; it estimates local Frobenius traces, not a full Jacobian matrix or a closed-loop guarantee.

Task	encoder	rollout	composed	alignment
TwoRoom	0.004	1.256	0.003	0.644
PushT	0.017	0.995	0.010	0.633
Reacher	0.003	0.887	0.006	1.739
Cube	0.023	1.079	0.028	1.042

Table 15: Threshold sensitivity for recovered-vs-fragile diagnostic separation. Entries count tasks for which recovered rows have lower median normalized ATR and higher median SMPR than fragile rows under the specified behavioral-label definition.

Recovery fraction	clean tolerance	ATR pass tasks	SMPR pass tasks	rows
0.70	3	4/4	4/4	4
0.70	5	4/4	4/4	4
0.70	10	4/4	4/4	4
0.80	3	4/4	4/4	4
0.80	5	4/4	4/4	4
0.80	10	4/4	4/4	4
0.90	3	4/4	4/4	4
0.90	5	4/4	4/4	4
0.90	10	4/4	4/4	4

C Additional Gaussian Evaluation Tables

These tables report the full observation-only Gaussian evaluation columns available for the three-training-seed sweep behind Figure 1. Each training-seed value first averages evaluation seeds 42/43/44; table cells report mean $\pm$ population std across training seeds 3072/3073/3074.

Table 16: Full three-training-seed LeWM observation-only Gaussian sweep evaluation table. Columns report clean evaluation and observation-only Gaussian evaluation at $\sigma = 0.03/0.05/0.08$ with a clean goal image. Values are mean $\pm$ population std across training seeds 3072/3073/3074; each training-seed value first averages evaluation seeds 42/43/44 with 100 trajectories per evaluation seed.

Task	training $\sigma_{\max}$	clean eval	obs-only $\sigma = 0.03$ eval	obs-only $\sigma = 0.05$ eval	obs-only $\sigma = 0.08$ eval
TwoRoom	0.00	94.67 $\pm$ 0.72	92.56 $\pm$ 1.23	81.33 $\pm$ 6.77	68.78 $\pm$ 2.45
TwoRoom	0.01	94.89 $\pm$ 1.10	95.00 $\pm$ 0.82	94.78 $\pm$ 1.77	93.44 $\pm$ 2.08
TwoRoom	0.02	95.78 $\pm$ 1.10	95.89 $\pm$ 1.29	95.22 $\pm$ 1.29	95.44 $\pm$ 1.57
TwoRoom	0.03	95.11 $\pm$ 0.87	95.00 $\pm$ 0.82	95.22 $\pm$ 0.83	95.11 $\pm$ 1.34
TwoRoom	0.04	95.89 $\pm$ 0.42	96.44 $\pm$ 0.96	95.78 $\pm$ 0.31	95.89 $\pm$ 0.57
TwoRoom	0.05	96.11 $\pm$ 1.50	96.00 $\pm$ 1.91	96.00 $\pm$ 1.52	95.89 $\pm$ 1.55
TwoRoom	0.06	96.00 $\pm$ 1.70	95.56 $\pm$ 1.81	95.56 $\pm$ 1.40	95.78 $\pm$ 1.55
TwoRoom	0.07	96.44 $\pm$ 0.63	96.33 $\pm$ 0.54	96.11 $\pm$ 0.68	96.44 $\pm$ 0.31
TwoRoom	0.08	97.22 $\pm$ 0.83	97.22 $\pm$ 0.68	97.33 $\pm$ 0.72	97.11 $\pm$ 0.57
PushT	0.00	81.78 $\pm$ 4.89	53.56 $\pm$ 12.91	19.22 $\pm$ 9.12	7.22 $\pm$ 5.81
PushT	0.01	85.89 $\pm$ 1.50	82.22 $\pm$ 3.03	70.33 $\pm$ 6.33	37.89 $\pm$ 12.37
PushT	0.02	87.33 $\pm$ 0.72	85.11 $\pm$ 1.13	82.89 $\pm$ 2.69	62.67 $\pm$ 11.98
PushT	0.03	89.67 $\pm$ 0.00	89.67 $\pm$ 0.27	88.78 $\pm$ 2.08	82.44 $\pm$ 1.66
PushT	0.04	89.11 $\pm$ 0.83	88.22 $\pm$ 1.13	87.33 $\pm$ 1.19	86.56 $\pm$ 0.16
PushT	0.05	84.00 $\pm$ 3.03	84.22 $\pm$ 2.53	82.56 $\pm$ 3.13	81.11 $\pm$ 3.03
PushT	0.06	88.22 $\pm$ 1.34	88.89 $\pm$ 1.03	87.33 $\pm$ 0.47	86.78 $\pm$ 0.63
PushT	0.07	86.44 $\pm$ 0.87	86.00 $\pm$ 1.19	84.22 $\pm$ 0.42	84.00 $\pm$ 0.27
PushT	0.08	86.89 $\pm$ 1.59	86.56 $\pm$ 2.20	84.67 $\pm$ 2.88	85.78 $\pm$ 3.45
Reacher	0.00	59.22 $\pm$ 1.03	42.56 $\pm$ 2.99	33.56 $\pm$ 1.77	18.22 $\pm$ 0.42
Reacher	0.01	62.67 $\pm$ 1.19	60.67 $\pm$ 1.09	56.00 $\pm$ 0.72	44.56 $\pm$ 2.57
Reacher	0.02	77.00 $\pm$ 6.18	76.44 $\pm$ 8.17	77.22 $\pm$ 6.68	70.33 $\pm$ 9.46
Reacher	0.03	78.22 $\pm$ 1.66	79.33 $\pm$ 0.72	79.11 $\pm$ 1.10	76.78 $\pm$ 4.17
Reacher	0.04	81.78 $\pm$ 1.59	82.00 $\pm$ 1.52	81.11 $\pm$ 2.62	80.56 $\pm$ 1.03
Reacher	0.05	76.44 $\pm$ 4.57	77.33 $\pm$ 5.76	77.56 $\pm$ 4.01	77.78 $\pm$ 5.67
Reacher	0.06	82.22 $\pm$ 2.79	79.33 $\pm$ 2.42	81.67 $\pm$ 1.63	80.67 $\pm$ 1.19
Reacher	0.07	82.11 $\pm$ 2.20	82.44 $\pm$ 1.29	81.44 $\pm$ 1.85	83.33 $\pm$ 1.09
Reacher	0.08	81.33 $\pm$ 2.33	82.56 $\pm$ 2.08	82.00 $\pm$ 1.78	81.56 $\pm$ 0.87
Cube	0.00	65.22 $\pm$ 1.10	63.00 $\pm$ 1.19	46.67 $\pm$ 1.96	43.11 $\pm$ 3.27
Cube	0.01	66.78 $\pm$ 1.81	67.00 $\pm$ 2.13	67.56 $\pm$ 2.27	48.78 $\pm$ 0.31
Cube	0.02	63.67 $\pm$ 2.62	61.44 $\pm$ 3.45	63.33 $\pm$ 1.96	56.67 $\pm$ 3.07
Cube	0.03	64.56 $\pm$ 1.66	63.44 $\pm$ 3.03	65.33 $\pm$ 0.94	66.00 $\pm$ 1.70
Cube	0.04	68.11 $\pm$ 1.03	65.11 $\pm$ 1.50	65.44 $\pm$ 0.57	64.11 $\pm$ 0.83
Cube	0.05	62.67 $\pm$ 3.40	63.33 $\pm$ 2.42	60.67 $\pm$ 3.09	63.56 $\pm$ 2.53
Cube	0.06	66.00 $\pm$ 1.19	65.89 $\pm$ 1.03	64.67 $\pm$ 1.09	64.89 $\pm$ 1.85
Cube	0.07	64.67 $\pm$ 2.16	64.67 $\pm$ 1.89	65.44 $\pm$ 2.57	65.22 $\pm$ 1.55
Cube	0.08	61.67 $\pm$ 3.57	61.78 $\pm$ 4.16	61.89 $\pm$ 2.45	62.56 $\pm$ 1.55